

Myra Cropper

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EDUCATION

Carnegie Mellon University: Pittsburgh, PA.

Master's of Science in Electrical and Computer Engineering - Artificial Intelligence.

December 2026

William & Mary: Williamsburg, VA.

Bachelor of Science in Computer Science, Minor in Math.

May 2025

FOUNDATIONAL SKILLS

- Deep learning architectures (CNN, ViT, LSTM) for perception, sensor fusion, and temporal modeling
- Embedded deployment on NVIDIA Jetson and Android; model optimization with TensorRT / ONNX
- ROS2, CAN bus communication (MCP2515), and real-time system/sim integration (Vehicles, Carla, MetaDrive)
- Distributed ML pipeline development with Kubeflow, KServe, Kafka, Docker, and Kubernetes
- Strong foundations in linear algebra, probability, and optimization

FRAMEWORKS

Languages: Python, Java, C, C++; **Libraries:** PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, ROS2, Hugging Face

Cloud & MLOps: Azure, AWS, Docker, Kubeflow, Kubernetes, KServe; **Databases & Storage:** PostgreSQL, Firebase, ChromaDB

PROJECTS

Comma_Hack6 Project

- Built a real-time autonomous driving inference pipeline integrating NVIDIA Alpamayo-R1, openpilot, Comma 3 simulation streams, and Ultra96-V2 FPGA acceleration, with 10 Hz trajectory planning with 110 ms end-to-end latency.
- Implemented FPGA-accelerated INT8 trajectory decoding for a diffusion-based driving model, achieving 11.46 ms inference latency
- Developed distributed inference infrastructure connecting camera streams, Alpamayo inference daemons, FPGA hardware, and openpilot planning modules through low-latency TCP/USB relay services.

Lidar Safety Validation

- Developed a safety-validation framework for NVIDIA's Alpamayo 1.5 in NVIDIA AlpaSim, combining LiDAR obstacle tracking, time-to-collision (TTC) estimation, physics-based trajectory arbitration across 6,450 validation timesteps and 42 autonomous driving scenarios.
- Reduced NVIDIA Alpamayo 1.5 collision frequency by 66.7% (14.3% to 4.8%) through a parallel safety-enforcer implementing emergency braking, stopping-distance validation, DBSCAN-based LiDAR obstacle clustering, low-latency trajectory override logic

RESEARCH EXPERIENCE

Drive Lab: Research Assistant, Carnegie Mellon: Pittsburgh, PA. (Targeting ICRA 2026 submission)

August 2025 - Present

- Designed a 4D radar simulation + detection pipeline: generated 4d radar point clouds in CARLA, built spherical+Doppler features, and trained RaDelft's Temporal ResNet achieving 67% Pd and 8.3% Pfa
- Validated a CARLA-based 4D radar simulation pipeline against the RaDelft Dataset, achieving JSD 0.0411, L1 distance 0.2158, KL divergences of 0.1755/0.1716, and Chamfer distances of 2.54–3.37 across simulated towns

Vehicle Computing Lab: Autonomous Vehicle Research, William & Mary: Williamsburg, VA.

Honors Thesis: "Multi-Sensor Fusion and V2V Communication in Autonomous Vehicles"

January 2024 – December 2025

- Fine-tuned EfficientNet model with LSTM with a 92% accuracy for detecting lane lines, used ROS2 to deploy on differential-drive robot
- Implemented YOLO models with 94% accuracy for detection/object tracking on traffic signs, traffic lights, and other vehicles
- Utilized Data Distributed Services (DSS) via ROS2 for multi-robot communication for robot parking and intersection crossing
- Formulated camera lidar fusion for distance-based object detection; modeled robot environment Octomapping in RVIZ

PROFESSIONAL EXPERIENCE

Isuzu Technical Center of America: Plymouth, MI

February 2026 – May 2026

AI & Autonomous Driving Intern (Logistics Hub Simulation)

- Built a logistic hub simulation control stack spanning 5 integrated components, NVIDIA Isaac Sim, MCP, gpt-oss:20b/Qwen 3.5, ROS 2 vehicle publishers, and a live monitoring dashboard
- Implemented 46 MCP tools for scene generation, vehicle control, semantic map tagging, dock occupancy detection, ROS 2 publishing
- Developed a ROS 2 vehicle bridge that published per-truck state including x/y/z, priority, task metadata, and fuel level
- Built a live truck-state dashboard to display vehicle name, position, priority, task duration, fuel level, and ROS topic in real time.

EverWatch – Booz Allen: Annapolis Junction, MD.

AI/ML/Cloud Computing Intern (AI Sensemaking Platform, DroneVLM)

June 2025 – August 2025

- Engineered a 5-agent LangGraph pipeline with PostgreSQL, SentenceTransformers, and locally hosted DeepSeek-R1 LLM, ingesting 6,700+ OSINT records from social media and news into a ChromaDB vector store for semantic search
- Integrated MCP to coordinate multi-agent collaboration, enabling contextual narrative synthesis across domain-specific sub agents with NER, sentiment analysis, and geolocation tools for detecting sentiment spikes
- Optimized RAG retrieval using LangGraph by developing a prompt-driven proxy agent dynamically routing queries to domain-specific subagents, and optimized queries via LLM, improving response relevance and reducing query latency by 75%
- Programmed an autonomous DJI Tello drone with SmolVLM, PyTorch, and OpenCV to scan environments for physical cybersecurity threats, operating grid-based navigation and real-time image analysis for threat detection and logging

